Supplementary Material for Edge-AI Road Surface Classification with Fuzzy-Adaptive Mixture-of-Experts

S1. Fuzzy Controller Specification

This section gives every value needed to rebuild the controller of Sec. III-E of the main paper. The same values are stored in the configuration file `fuzzy_v2_r3.json` in our code release. The controller is implemented with scikit-fuzzy 0.5.0.

A. Inputs

The controller has three inputs. All three are computed from the validation pass of the previous epoch.

- BS: the balance score of Eqs. (7)–(8) of the main paper. Expert usage is the class-prior-weighted share of top-2 selections on the validation set. We use $\beta = 0.8$.
- T_{fast}: the accuracy trend of Eq. (9) with window $W_f = 3$. The accuracy is the validation A_π.
- T_{slow}: the same trend with window $W_s = 10$.

Each input is clipped to its universe before fuzzification. A trend is set to 0 while fewer than $W + 1$ epochs of history exist.

B. Membership Functions

Table S1 lists all membership functions. A triangle is written $\mathrm{tri}(a, b, c)$ with its peak at b. A trapezoid is written $\mathrm{trap}(a, b, c, d)$ with its plateau on $[b, c]$. Each universe is discretized with the listed step.

The input breakpoints follow the distribution of the three signals in the pilot runs of Fixed MoE ($\lambda = 1.5$, $\sigma = 0.4$, $\tau = 0.5$, 50 epochs, three pilot seeds). For BS, the 0.5-membership crossings of Unstable and Perfect sit near the 5th and 85th percentiles, and the Normal peak sits at the median (86.6). The Degrading term of T_{fast} lies outside the epoch-to-epoch noise of the validation accuracy, so R10 fires only on a sharp drop.

The output universes equal the ranges found acceptable for Fixed MoE in the pilot study (Sec. S2). The Medium terms of λ and σ have their centroids at 1.5 and 0.4, the calibrated Fixed MoE values.

C. Rules and Inference

The rule base is Table I of the main paper. Rules R1–R9 cover every combination of BS and T_{slow}. R10 depends on T_{fast} only. All conditions are joined by AND.

Inference uses Mamdani's method: min for AND, min implication, and max aggregation. Each output is defuzzified by its centroid.

D. Update Procedure

Epochs are numbered from 1. The controller acts at the start of epoch t, before the training pass.

1) Epochs 1–3 are the learning-rate warm-up. The controller is inactive, and the coefficients keep their initial values $\tau_0 = 0.5$, $\lambda_0 = 1.5$, and $\sigma_0 = 0.4$.
2) Epochs 4–11 are the early mode. T_{slow} has no history yet, so only R10 is evaluated. When R10 does not fire, the coefficients keep their previous values.
3) From epoch 12, all ten rules are evaluated.
4) The temperature is updated incrementally:
$$\tau_t = \mathrm{clip}(\tau_{t-1} + \Delta\tau_t,\ 0.25,\ 1.0). \tag{S1}$$
5) The weight and the noise follow an exponential moving average of the rule outputs λ_t^{out} and σ_t^{out}:
$$\lambda_t = 0.8\,\lambda_{t-1} + 0.2\,\lambda_t^{\text{out}}, \quad \sigma_t = 0.8\,\sigma_{t-1} + 0.2\,\sigma_t^{\text{out}}. \tag{S2}$$
6) If inference fails, the coefficients keep their previous values. This did not occur in any of the 12 FuzzyAdp-MoE runs of the main experiments.

The noise σ is added to the cosine scores during training only. At inference the router uses the temperature τ of the selected checkpoint and no noise.

E. Linear Schedule Baseline

Schedule MoE uses the same router, loss, and output ranges, but does not read any training signal. During epochs 1–3 it holds the start values. From epoch 4 to epoch 50 the three coefficients move linearly: λ from 4.5 to 0.5, σ from 0.8 to 0.1, and τ from 0.5 to 0.25. This is the nominal trajectory of the rule base: strong balancing and exploration early, weak balancing and sharp routing late.

F. Design Rounds

The rule base was designed in three rounds on the pilot seeds. The test split was not used. Table S2 lists the changes. Round 3 is the version used in all reported experiments.

S2. Pilot Study

The pilot study selected every setting of the main experiments. It used seeds 0–2, which are not reused in the main experiments (seeds 3–5), and the validation subset of 5,810 images. The test split was not used. Each stage fixed the choices of the earlier stages, and the stages are reported in

TABLE S1
MEMBERSHIP FUNCTIONS OF THE FUZZY CONTROLLER.

Variable	Universe (step)	Term	Membership function
BS	$[50, 100]$ (0.5)	Unstable	$\mathrm{trap}(50, 50, 74, 80)$
		Normal	$\mathrm{tri}(74, 86.5, 94)$
		Perfect	$\mathrm{trap}(87, 93, 100, 100)$
T_{fast}	$[-0.04, 0.04]$ (0.0005)	Degrading	$\mathrm{trap}(-0.04, -0.04, -0.025, -0.018)$
T_{slow}	$[-0.03, 0.05]$ (0.0005)	Degrading	$\mathrm{trap}(-0.03, -0.03, -0.008, 0)$
		Stable	$\mathrm{tri}(-0.006, 0.003, 0.012)$
		Improving	$\mathrm{trap}(0.006, 0.015, 0.05, 0.05)$
$\Delta\tau$	$[-0.075, 0.075]$ (0.0005)	Decrease	$\mathrm{tri}(-0.075, -0.045, 0)$
		Keep	$\mathrm{tri}(-0.015, 0, 0.015)$
		Increase	$\mathrm{tri}(0, 0.045, 0.075)$
λ	$[0.5, 4.5]$ (0.01)	Low	$\mathrm{trap}(0.5, 0.5, 0.7, 1.2)$
		Medium	$\mathrm{tri}(0.5, 1.5, 2.5)$
		High	$\mathrm{trap}(2.5, 3.8, 4.5, 4.5)$
σ	$[0.1, 0.8]$ (0.001)	Low	$\mathrm{trap}(0.1, 0.1, 0.15, 0.3)$
		Medium	$\mathrm{tri}(0.15, 0.4, 0.65)$
		High	$\mathrm{trap}(0.5, 0.65, 0.8, 0.8)$

TABLE S2
DESIGN ROUNDS OF THE FUZZY CONTROLLER (PILOT SEEDS 0–2).

Round	Change from the previous round	ΔA_π vs. Fixed MoE (pp, val.)
1	Initial design: rules R1–R10, breakpoints from the pilot distributions	-0.61
2	Moving average (0.8) on λ and σ; R10 breakpoints moved out of the noise range; $\Delta\tau$ range scaled by 1.5	-0.84
3	Output σ of R5, R8, and R9 raised from Low to Medium	-0.23

the order in which they condition each other. All values are the best validation A_π of a run unless stated otherwise. The run logs and the summary scripts are part of the code release.

A. Protocol and Decision Rules

Starting settings. Augmentation B (horizontal flip, scale 0.9–1.1, mild color jitter), peak learning rate 10^{-3}, 70 epochs, weight decay 10^{-4}, and $(\lambda_0, \sigma_0, \tau_0) = (1.5, 0.2, 0.5)$. The routing values were provisional: they came from a 20-epoch calibration under an earlier router and were recalibrated in Sec. S2-D.

Threshold. $X = \max(0.3 \text{ pp}, r)$, where r is the range over three seeds of the best validation A_π of Fixed MoE at the starting settings. The three runs gave 69.93%, 70.43%, and 70.08%, so $r = 0.50$ pp and $X = 0.50$ pp. An earlier measurement under the previous router with the auxiliary-loss warm-up gave 0.43 pp. Every decision below is the same under either value.

Rules. Candidates were compared with the default of their stage. In a single-seed comparison, a candidate within X of the best value ties, and a tie keeps the default (for the epoch count, the shortest candidate). A candidate that beat the default by at least X on seed 0 was rerun on seeds 1 and 2 and adopted only if the seed-paired mean difference $\bar{d}$ reached $+X$. A shorter epoch count was kept if $\bar{d} \geq -X$. A run was excluded if its lowest class-prior-weighted expert usage over the last 10 epochs fell below 5% in any seed. The standard error of a three-seed paired mean was 0.14–0.42 pp in the pilot, the same order as X, so a tie means that no difference was found at this precision.

B. Structure Checks

Table S3 reports the two structural checks, both at the starting settings with three seeds.

Auxiliary-loss warm-up. The submitted version multiplied λ by 0.1 during the first third of training. Removing the warm-up changed A_π by $+0.03$ pp, and no seed was excluded. With the full λ from the first epoch, the auxiliary loss stays near its minimum of 1.0 from the start. The warm-up was removed: it had no literature basis, it left the controller without control over λ for a third of training, and Aux-free MoE has no such coefficient. A property of the loss became visible here: $\mathcal{L}_{\text{aux}}$ counts both top-2 positions, so an expert that is selected mainly as the second choice has a normal f_i. Neither the loss nor BS sees this case, which is why the main paper also reports the lowest first-choice (top-1) share.

Router. The submitted router (two 3×3 convolutions with batch normalization, ReLU, and dropout, followed by a cosine head; 371,552 parameters, 18.06 M multiply–accumulates per sample) was replaced by the cosine router of Sec. III-B (3,168 parameters). The cosine router was $+0.31$ pp better on the seed-paired mean. With the convolutional router, the weakest expert's top-1 share over the last 10 epochs was 4.1%, 0.7%,

and 0.0% on the three seeds while its usage stayed near 16%; with the cosine router it was at least 12.4%.

C. Training Hyperparameters

Augmentation, peak learning rate, epoch count, and weight decay were selected in this order, each under the choices before it (Table S4). Fixed MoE was the selecting model; the monolithic network was run alongside for the augmentation and the learning rate, and its choices agreed. The epoch count was placed before the weight decay because the effective decay of AdamW depends on the number of steps.

Three decisions needed the three-seed confirmation of Sec. S2-A.

- **50 vs. 70 epochs.** The three candidates tied on seed 0, so the shortest was taken to confirmation. Paired differences 50 minus 70: -0.29, -0.43, -0.10 pp, $\bar{d} = -0.27$ pp $\geq -X$. Fifty epochs were adopted. The training accuracy was about 0.999 at the end of both lengths.
- **Weight decay of Fixed MoE.** 0.05 was 0.91 pp above 10^{-4} on seed 0. Paired differences: $+0.91$, $+0.70$, -0.40 pp, $\bar{d} = +0.40$ pp $< X$. The default 10^{-4} was kept.
- **Weight decay of the monolithic network.** 0.01 was 1.71 pp above 10^{-4} on seed 0. Paired differences: $+1.71$, $+1.45$, $+0.44$ pp, $\bar{d} = +1.20$ pp $\geq X$. The monolithic network uses 0.01. This confirmation was added after the seed-0 result, because forcing the MoE value on the monolithic network would have cost it 1.71 pp on that seed and biased the comparison toward the MoE.

Router learning rate. The router shares the learning rate of the backbone. The training diagnostics of the 50-epoch runs showed no reason for a separate value: after epoch 10, the relative update of the expert embeddings was 1.7–3.1 times that of the expert weights and stayed there; the loss-scaler skipped 13–16 steps per run, consistent with dynamic loss scaling; and the per-epoch change of the top-1 assignment followed the learning rate (Spearman correlation 0.93–0.97 across six runs).

D. Calibration of Fixed MoE

With the training settings fixed (50 epochs, weight decay 10^{-4}), λ, σ, and τ were varied one at a time around $(1.5, 0.2, 0.5)$ with three seeds each (Table S5). A value was adopted if $\bar{d} \geq +X$; it is acceptable if $\bar{d} \geq -X$ and no seed is excluded. Only $\sigma = 0.4$ qualified ($+0.59$ pp). No candidate was excluded. The acceptable values are $\lambda \in [0.5, 4.5]$, $\sigma \in [0.1, 0.8]$, and $\tau \in [0.25, 1.0]$; they are the output universes of the controller and the end points of the linear schedule. The $\sigma = 0.4$ decision exceeded X by 0.08 pp. For the record: with X equal to the range of the 50-epoch center (0.36 pp), $\lambda = 0.5$ and $\tau = 0.25$ would also have qualified; X was fixed in advance and was not changed.

Re-check at $\sigma_0 = 0.4$. Because σ_0 changed, the choices most likely to depend on it were re-checked with $\sigma_0 = 0.4$ as the baseline (Table S6). All passed. $\sigma = 0.8$ tied with 0.4 and was not excluded, so the acceptable range of σ was extended to 0.8; no larger value was tested. Fifty epochs were better than 70 on all three seeds. Weight decay 0.05 gave $\bar{d} = +0.37$ pp, close to its value at $\sigma_0 = 0.2$, and again below X.

E. Aux-free MoE

Aux-free MoE follows DeepSeek-V3: the affinity is $\mathrm{sigmoid}(s_i/\tau)$, the top-2 is selected on affinity plus a per-expert bias, the bias is used for selection only, and after every step the bias of an over-loaded expert is decreased by γ and that of an under-loaded expert increased by γ. There is no routing noise and no auxiliary loss. Two differences from the original: the scores are cosine scores, so τ is kept (the value of Fixed MoE, 0.5), and γ is constant to the end.

Its only free parameter, γ, was checked at 10^{-4}, 10^{-3} (the published value), and 10^{-2} (Table S7). At $\gamma = 10^{-3}$, no seed collapsed. $\gamma = 10^{-4}$ was 1.38 pp better on seed 0 and went to confirmation: seed 1 was excluded (lowest usage 0.01%), and all three seeds collapsed in mid-training (29, 43, and 29 epochs below 5%, the last of them epochs 36, 45, and 38). The seed-0 gain came from a checkpoint after the late recovery and does not show that the smaller γ is better. $\gamma = 10^{-2}$ tied on seed 0. The published value was kept. At $\gamma = 10^{-3}$, the bias changed the top-2 of 84%, 18%, and 75% of the validation images in the first epoch and of 8%, 12%, and 16% on average over the last 10 epochs.

F. Controller Input Distribution

The input breakpoints of Sec. S1 were set from the distribution of the three inputs in the Fixed MoE runs with $\sigma_0 = 0.4$ (three seeds, epochs 4–50; Table S8). BS fell below 70 in only two epochs; its minimum was 58.0. The accuracy trend is dominated by validation noise: the standard deviation of the epoch-to-epoch change of validation A_π was 1.28 pp over the whole run and 0.57 pp over epochs 36–50, and the correlation of T_{fast} between adjacent epochs was 0.14. The design therefore uses T_{fast} only for the sharp-drop rule R10, with its breakpoints outside the noise band, and T_{slow} with $W_s = 10$ for the state rules. An open-loop replay of the submitted six-rule base on the same runs kept τ at its upper limit and σ near 0.12 throughout; this motivated the redesign.

G. Selection of the Poor Starting Points

The three starting points of Sec. IV-D.3 were found by a probe of Fixed MoE on seed 0 (Table S9). The first batch moved one parameter at a time one grid step outside its acceptable range ($\sigma_0 = 0$ instead of 0.05, as the natural value when noise is not used). None of the six candidates lost balance, and only $\lambda_0 = 13.5$ was worse than the reference by more than X. A second batch, added after this result, removed the balancing loss ($\lambda_0 = 0$), moved λ_0 and σ_0 two steps outside, and combined the three parameters in the collapse direction, in the flattening direction, and in a mixed direction.

A start was *poor* if (1) the exclusion rule triggered, (2) more than five epochs after epoch 10 had usage below 5%, or (3) A_π was more than X below the reference (70.11%). The selection order was written before the second batch ran: usage criteria before the accuracy criterion; single-parameter

TABLE S3
STRUCTURE CHECKS AT THE STARTING SETTINGS (BEST VALIDATION A_π, %; SEEDS 0 / 1 / 2).

Check	**Default**	**Candidate**	d_k **(pp)**	$\bar{d}$ **(pp)**
Warm-up kept vs. removed	69.87 / 70.17 / 70.30	69.93 / 70.43 / 70.08	$+0.06$ / $+0.27$ / -0.22	$+0.03$
Conv. router vs. cosine router	69.88 / 69.84 / 69.77	69.93 / 70.43 / 70.08	$+0.05$ / $+0.59$ / $+0.31$	$+0.31$

TABLE S4
TRAINING HYPERPARAMETERS (SEED 0, BEST VALIDATION A_π, %). DEFAULTS IN BOLD.

Step	**Candidates**	**Fixed MoE**	**Monolithic**	**Choice**
Augmentation (70 ep.)	A / **B**	67.63 / 69.93	67.76 / 69.96	B
Peak learning rate (70 ep.)	10^{-4} / $3{\cdot}10^{-4}$ / $\mathbf{10^{-3}}$ / $3{\cdot}10^{-3}$	68.13 / 69.67 / 69.93 / 69.13	68.80 / 68.93 / 69.96 / 69.88	10^{-3}
Epochs	50 / **70** / 100	69.64 / 69.93 / 69.62	—	50
Weight decay (50 ep.)	$\mathbf{10^{-4}}$ / 0.01 / 0.05 / 0.1	69.64 / 69.58 / 70.55 / 69.22	69.02 / 70.73 / 70.68 / 68.94	MoE 10^{-4}, monolithic 0.01

TABLE S5
CALIBRATION OF FIXED MOE (50 EPOCHS, BEST VALIDATION A_π, %; SEEDS 0 / 1 / 2). USAGE IS THE LOWEST CLASS-PRIOR-WEIGHTED EXPERT USAGE OVER THE LAST 10 EPOCHS.

Configuration	A_π	d_k **(pp)**	$\bar{d}$ **(SE)**	**Usage (%)**	**NMI**
Center $(1.5, 0.2, 0.5)$	69.64 / 70.00 / 69.98	—	—	17.2 / 14.8 / 16.9	0.242 / 0.209 / 0.206
$\lambda = 0.5$	70.27 / 70.58 / 70.09	$+0.63$ / $+0.58$ / $+0.12$	$+0.44$ (0.16)	16.6 / 15.5 / 15.6	0.230 / 0.224 / 0.253
$\lambda = 4.5$	69.72 / 70.29 / 68.94	$+0.08$ / $+0.29$ / -1.04	-0.22 (0.41)	17.3 / 15.4 / 16.6	0.191 / 0.191 / 0.190
$\sigma = 0.1$	69.51 / 70.77 / 70.04	-0.13 / $+0.77$ / $+0.06$	$+0.23$ (0.27)	16.7 / 14.1 / 16.1	0.238 / 0.201 / 0.230
$\sigma = 0.4$ (adopted)	70.11 / 70.84 / 70.42	$+0.47$ / $+0.84$ / $+0.45$	$+\mathbf{0.59}$ (0.13)	17.2 / 15.2 / 17.0	0.261 / 0.222 / 0.261
$\tau = 0.25$	70.18 / 70.45 / 70.19	$+0.55$ / $+0.45$ / $+0.21$	$+0.40$ (0.10)	18.0 / 15.2 / 16.1	0.234 / 0.227 / 0.270
$\tau = 1.0$	69.00 / 70.59 / 69.88	-0.64 / $+0.59$ / -0.09	-0.05 (0.36)	17.5 / 14.2 / 15.5	0.205 / 0.208 / 0.208
$\sigma = 0.8$ (vs. center)	70.43 / 70.23 / 70.34	$+0.79$ / $+0.23$ / $+0.36$	$+0.46$	16.9 / 11.6 / 14.0	0.276 / 0.255 / 0.276

TABLE S6
RE-CHECK OF EARLIER CHOICES AT $\sigma_0 = 0.4$ (SEED-PAIRED DIFFERENCES, PP).

Comparison	d_k **(seeds 0 / 1 / 2)**	$\bar{d}$	**Outcome**
$\sigma = 0.8$ vs. 0.4	$+0.32$ / -0.61 / -0.09	-0.13	keep 0.4; range extended to 0.8
50 vs. 70 epochs	$+1.00$ / $+0.59$ / $+0.96$	$+0.85$	keep 50
Weight decay 0.05 vs. 10^{-4}	$+0.83$ / $+0.59$ / -0.30	$+0.37$	keep 10^{-4}
Learning rate $3{\cdot}10^{-4}$ vs. 10^{-3}	-0.59 (seed 0)	—	keep 10^{-3}
Learning rate $3{\cdot}10^{-3}$ vs. 10^{-3}	-0.35 (seed 0)	—	keep 10^{-3}

TABLE S7
BIAS STEP γ OF AUX-FREE MOE (BEST VALIDATION A_π, %; SEEDS 0 / 1 / 2).

γ	A_π	d_k **vs.** 10^{-3} **(pp)**	**Usage, last 10 ep. (%)**	**Outcome**
10^{-3}	69.54 / 69.72 / 70.40	—	16.5 / 15.5 / 17.0	kept
10^{-4}	70.92 / 69.99 / 69.35	$+1.38$ / $+0.27$ / -1.05 ($\bar{d} = +0.20$)	16.4 / 0.01 / 17.5	seed 1 excluded
10^{-2}	70.02 (seed 0)	$+0.48$	—	tie

TABLE S8
PERCENTILES OF THE CONTROLLER INPUTS IN THE FIXED MOE PILOT RUNS ($\sigma_0 = 0.4$, EPOCHS 4–50, THREE SEEDS).

Input	p5	p50	p95
BS	77.3	86.6	92.2
T_{fast} ($W_f = 3$)	-0.0096	0.0026	0.0216
T_{slow} ($W_s = 10$)	-0.0031	0.0067	0.0209

starts before same-direction combinations before mixed combinations; smaller deviations first. The first three were $\lambda_0 = 0$, the collapse combination $(0, 0, 0.125)$, and the flattening combination $(40.5, 3.2, 2.0)$. Fixed MoE was then run on seeds 1 and 2 at these starts; the usage criteria triggered on 3 of 3 seeds at the first two, and the flattening start was -0.70 pp below the baseline start on the seed-paired mean. All three were confirmed as poor before FuzzyAdp-MoE was run.

Two observations from the probe carry over to the main experiments. Without the balancing loss, the routing collapsed from the first epochs regardless of σ_0 and τ_0: at $(0, 0, 0.125)$ and $(0, 3.2, 2.0)$, the same two experts received every image from epoch 2 on. Collapse did not reduce validation accuracy on seed 0: the three $\lambda_0 = 0$ starts were within X of the reference.

H. Tuning Effort per Method

Table S10 lists what was tuned for each method. The effort is not equal. Fixed MoE received the full calibration. FuzzyAdp-MoE was designed in three rounds on the pilot seeds; its rounds 2 and 3 were decided after seeing the validation accuracy of the previous round (Table S2), which is one layer of selection more than Fixed MoE had.

S3. ROBUSTNESS TO POOR STARTING POINTS

Table S11 gives the per-seed results behind the robustness table of the main paper (Sec. IV-D). Accuracy is the test A_π of the checkpoint with the best validation A_π. Usage is the lowest class-prior-weighted expert usage on the validation set, taken per epoch from the training history. The recovery epoch is the first epoch after which usage stays at or above 5%.

Collapse and recovery. At the two starts without the balancing loss, the weakest expert of Fixed MoE received 1–5% of the validation images in epoch 1 and none from epoch 3 on, and two experts received almost all images for the rest of training. Seed 4 at $(0, 0.4, 0.5)$ is the one partial exception: its lowest usage rose to 8.5% and 9.5% in epochs 49 and 50, while one expert still received 50% of the images. Its lowest usage over the last 10 epochs is 0.0%, so it is collapsed under the pre-registered criterion. FuzzyAdp-MoE starts from the same collapsed state, because the controller cannot act during the warm-up and reads only T_{fast} during the early mode. Recovery then came from two paths. At $(0, 0.4, 0.5)$, R10 fired once or twice on seeds 3 and 5 between epochs 5 and 11, raising λ to 0.76–1.35, and usage was back above 5% at epochs 6 and 7. On seed 4, R10 did not fire; λ started to rise only at epoch 12, when all rules became active, and usage returned above 5% at epoch 29. At $(0, 0, 0.125)$, all three seeds recovered at epochs 10–12. In the full mode, BS was Unstable and R1–R3 raised all three coefficients. At $(0, 0.4, 0.5)$, λ peaked at 3.1–3.6 and σ at 0.58–0.64, and τ reached 0.94–1.0. At $(0, 0, 0.125)$, λ peaked at 1.7–3.5 and settled at 0.9–1.5, σ rose from 0 to 0.42–0.46 at epoch 50 (peak 0.55–0.59), and τ reached 0.25–0.95. At the flattening start, λ decayed under the moving average from 40.5 to 6.5–7.6 at epoch 21 and to 1.0–2.7 at epoch 50, σ from 3.2 to 0.43–0.53, and τ was clipped to 1.0 from the first update. Fixed MoE did not collapse at this start; the comparison there is between an extreme but balanced coefficient set and the calibrated range.

Accuracy. The seed-paired differences FuzzyAdp-MoE minus Fixed MoE are -0.63, $+0.51$, -1.69 pp (mean -0.60) at $(0, 0.4, 0.5)$; $+0.31$, $+0.05$, -0.42 pp (mean -0.02) at $(0, 0, 0.125)$; and $+2.86$, -1.22, -1.13 pp (mean $+0.17$) at $(40.5, 3.2, 2.0)$. For description only: the collapsed Fixed MoE runs are not less accurate than Fixed MoE at the calibrated start (seed-paired means $+0.31$ and $+0.40$ pp at the two starts without the balancing loss, both within X), and FuzzyAdp-MoE at the poor starts is -0.90, -0.22, and -0.79 pp from FuzzyAdp-MoE at the calibrated start. On this task, routing collapse onto two experts does not cost test accuracy, so recovering balanced usage does not raise it.

Experts after recovery. The dense evaluation of Sec. S4 was also run on the six recovered FuzzyAdp-MoE models and the three from the flattening start. After recovery from $(0, 0, 0.125)$, masking the top-1 expert costs $+0.62$, $+0.26$, and 0.00 pp, with mean CKA 0.61, 0.59, 0.59 and agreement of the two selected experts 0.815, 0.829, 0.832: the same level as Fixed MoE at $\sigma = 0.4$ and far from Fixed MoE at $\sigma = 0$ (13–15 pp, Table S15). The recovery path raises σ to 0.4 and above, and the recovered models have no functional specialization. The models from the flattening start, whose σ stayed above 1.6 until epoch 15, are the most homogeneous of all evaluated runs (CKA 0.81, 0.78, 0.85; agreement 0.895, 0.875, 0.902).

The same three starts were tested in the pilot on seeds 0–2 with the same outcome on balance (9 of 9 FuzzyAdp-MoE runs recovered, 6 of 6 Fixed MoE runs at the two starts without the balancing loss did not); the validation seed-paired means were $+0.16$, -0.32, and $+0.60$ pp.

S4. FUNCTIONAL ANALYSIS OF THE EXPERTS

A. Measures

All measures are computed on the validation set from the checkpoint with the best validation A_π, with the trained router and no noise. A *dense* pass first records the logits of all five experts for every image, so that any selection of experts can be re-assembled; re-assembling the top-2 mixture reproduces the reported A_π of the main runs exactly.

- **A_π drop, top-1 (top-2) expert masked (d_1, d_2):** the expert(s) chosen by the router are removed and the image is re-routed among the remaining ones. Near 0 when the experts are interchangeable.

TABLE S9
PROBE OF POOR STARTING POINTS (FIXED MOE, SEED 0, 50 EPOCHS; REFERENCE $(1.5, 0.4, 0.5)$ = 70.11%).

Start $(\lambda_0, \sigma_0, \tau_0)$	A_π **(%)**	**vs. ref. (pp)**	**Usage, last 10 ep. (%)**	**Epochs** $< 5\%$ **after ep. 10**	**Top-1 share, last 10 ep. (%)**	**Poor**
$(0.17, 0.4, 0.5)$	69.63	-0.48	17.8	0	8.7	no
$(13.5, 0.4, 0.5)$	69.45	-0.66	14.7	0	8.0	(3)
$(1.5, 0, 0.5)$	69.73	-0.38	17.0	0	14.9	no
$(1.5, 1.6, 0.5)$	70.37	$+0.26$	12.2	0	9.5	no
$(1.5, 0.4, 0.125)$	69.90	-0.21	18.3	0	10.8	no
$(1.5, 0.4, 2.0)$	69.77	-0.34	16.2	0	8.8	no
$(0, 0.4, 0.5)$	70.46	$+0.34$	0.02	38	0.0	(1)(2), selected
$(40.5, 0.4, 0.5)$	70.13	$+0.02$	16.3	0	11.5	no
$(1.5, 3.2, 0.5)$	69.29	-0.82	8.4	0	11.6	(3)
$(0, 0, 0.125)$	70.21	$+0.09$	0.0	40	0.0	(1)(2), selected
$(40.5, 3.2, 2.0)$	69.25	-0.86	3.4	18	5.5	(1)(2)(3), selected
$(0, 3.2, 2.0)$	69.84	-0.27	0.0	40	0.0	(1)(2)

TABLE S10
TUNING EFFORT PER METHOD IN THE PILOT STUDY.

Method	**What was tuned**
Monolithic	Augmentation and learning rate taken from the joint screening; weight decay confirmed on three seeds; epoch count not compared separately.
Fixed MoE	Augmentation, learning rate, epochs, weight decay (seed 0 plus confirmations); λ_0, σ_0, τ_0 one at a time with three seeds each; re-check at $\sigma_0 = 0.4$.
Schedule MoE	No tuning. End points are the acceptable ranges of Fixed MoE; the trajectory is the nominal one of the rule base.
Aux-free MoE	γ checked at three values; τ taken from Fixed MoE; no other adjustment.
FuzzyAdp-MoE	Output universes from the calibration; input breakpoints from the pilot distributions; rule base designed from the variable meanings, then revised twice on the pilot seeds (Table S2).

- **Gain of routing over 2 random experts:** A_π of the routed mixture minus A_π when two experts are drawn at random (expected value over all pairs).
- **Specialization gain:** for each class, the recall of its dominant expert (the expert most often ranked first on that class) minus the mean recall of the other experts on that class, weighted by the class prior.
- **Mean CKA:** mean linear centered kernel alignment between the logits of every pair of experts; higher means more similar.
- **Disagreement:** fraction of images on which two experts (averaged over pairs) predict different classes.
- **Agreement of the selected experts:** fraction of images on which the two routed experts predict the same class.
- **Class–expert NMI:** normalized mutual information $I(C;E)/H(E)$ between the class and the top-1 expert assignment, class-prior-weighted.
- **Top-1-only drop:** A_π of the top-2 mixture minus A_π when only the top-1 expert is run (the deployment variant of the main paper's latency table).
- **Any-of-5:** fraction of images that at least one of the five experts classifies correctly; an upper bound for a perfect router.

B. Main Experiments

Table S12 gives all routing and functional statistics behind Table III and Fig. 2 of the main paper, with the range over the three seeds. The NMI and assignment-similarity rows use the last checkpoint; Fig. 2 and Tables S13 and S14 use the selected checkpoint.

Tables S13 and S14 list every measure for the 12 MoE runs of the main experiments. The three groups do not overlap on d_1, routing gain, CKA, disagreement, or agreement, and their order follows the routing noise during the first 11 epochs, when the class-to-expert assignment forms: Aux-free MoE ($\sigma = 0$) has the most specialized experts, Fixed MoE and FuzzyAdp-MoE ($\sigma = 0.4$, rising to 0.48–0.55 by epoch 15 in FuzzyAdp-MoE) are in the middle, and Schedule MoE (σ from 0.8 to 0.69 over epochs 4–11, then down to 0.1) has the most similar experts, although its final σ is the lowest. NMI is 0.22–0.29 for all four methods. Specialization makes the experts more diverse (any-of-5 0.79–0.80 for Aux-free MoE against 0.73–0.75 for the others) but the router does not turn this into accuracy, and it makes the top-1-only variant more costly (0.8–1.9 pp against 0.0–0.7 pp).

C. Pilot Sweep over the Routing Noise

Tables S15 and S16 give the same measures for the Fixed MoE pilot runs (seeds 0–2) at $\sigma \in \{0, 0.1, 0.2, 0.4, 0.8\}$, for the single-seed τ and λ variations of the calibration, and for the pilot runs of the three other MoE methods. Specialization decreases monotonically with σ: d_1 is 12.7–15.2 pp at $\sigma = 0$, 4.5–7.4 pp at 0.1, 0.4–1.1 pp at 0.4, and 0.2–0.6 pp at 0.8. The three-seed groups at 0, 0.1, and 0.4 do not overlap on d_1, routing gain, or agreement; the groups at 0.4 and 0.8 are separated by CKA (0.47–0.59 against 0.71–0.75) and agreement (0.83–0.85 against 0.88–0.90). All runs of the four

TABLE S11
POOR STARTING POINTS, PER SEED. TEST A_π (%), BEST EPOCH, NUMBER OF EPOCHS WITH USAGE BELOW 5%, RECOVERY EPOCH, LOWEST USAGE OVER THE LAST 10 EPOCHS (%), AND THE CONTROLLER COEFFICIENTS AT EPOCH 50.

Start	**Model**	**Seed**	**Test A_π**	**Best ep.**	**Ep. $< 5\%$**	**Recovery ep.**	**Usage, last 10**	**(λ, σ, τ) at ep. 50**
$(0, 0.4, 0.5)$	Fixed MoE	3	58.86	40	50	—	0.0	$(0, 0.40, 0.50)$
		4	58.97	39	47	—	0.0	$(0, 0.40, 0.50)$
		5	59.21	43	50	—	0.0	$(0, 0.40, 0.50)$
	FuzzyAdp-MoE	3	58.23	46	5	6	16.4	$(0.99, 0.40, 0.94)$
		4	59.47	50	18	29	12.0	$(3.49, 0.62, 1.00)$
		5	57.51	42	6	7	12.9	$(3.60, 0.64, 1.00)$
$(0, 0, 0.125)$	Fixed MoE	3	58.78	47	50	—	0.0	$(0, 0, 0.125)$
		4	58.73	50	50	—	0.0	$(0, 0, 0.125)$
		5	59.80	45	50	—	0.0	$(0, 0, 0.125)$
	FuzzyAdp-MoE	3	59.09	47	11	12	15.8	$(1.23, 0.45, 0.74)$
		4	58.78	39	9	10	16.2	$(0.89, 0.42, 0.25)$
		5	59.38	46	11	12	15.7	$(1.54, 0.46, 0.95)$
$(40.5, 3.2, 2.0)$	Fixed MoE	3	55.60	32	1	19	13.3	$(40.5, 3.2, 2.0)$
		4	59.27	46	1	3	12.6	$(40.5, 3.2, 2.0)$
		5	60.16	35	0	—	15.0	$(40.5, 3.2, 2.0)$
	FuzzyAdp-MoE	3	58.46	41	1	15	15.5	$(1.42, 0.44, 0.99)$
		4	58.05	33	1	3	14.6	$(2.75, 0.53, 1.00)$
		5	59.03	43	0	—	15.6	$(1.04, 0.43, 0.75)$

TABLE S12
ROUTING AND FUNCTIONAL STATISTICS OF THE MOE VARIANTS, MEAN (MIN–MAX) OVER THREE SEEDS (VALIDATION SET).

Metric	**Fixed**	**Schedule**	**Aux-free**	**FuzzyAdp**
Min. expert usage, last 10 epochs (%)	15.7 (15.1–16.4)	17.8 (17.6–18.3)	17.0 (15.3–17.9)	16.6 (16.1–17.3)
Min. top-1 share, last 10 epochs (%)	10.9 (7.9–13.4)	7.4 (3.5–11.6)	9.2 (7.5–10.2)	11.9 (9.4–13.3)
Class–expert NMI	0.233 (0.232–0.235)	0.253 (0.229–0.268)	0.232 (0.216–0.242)	0.250 (0.236–0.260)
Max. assignment similarity of two experts	0.347 (0.106–0.594)	0.900 (0.855–0.937)	0.406 (0.227–0.558)	0.377 (0.165–0.750)
A_π drop, top-1 expert masked (pp)	0.42 (0.23–0.74)	0.04 (−0.37–0.59)	**6.05** (5.18–6.56)	0.59 (0.20–1.08)
Gain of routing over 2 random experts (pp)	0.55 (0.32–0.85)	0.15 (−0.06–0.35)	**4.47** (3.83–5.27)	0.58 (0.43–0.76)
Mean CKA between experts	0.50 (0.47–0.52)	0.70 (0.68–0.72)	**0.30** (0.25–0.32)	0.55 (0.51–0.59)
Prediction disagreement of selected experts	0.19 (0.18–0.20)	0.16 (0.14–0.18)	**0.44** (0.40–0.46)	0.17 (0.16–0.18)

Note: All statistics are on the validation set. The first four rows describe routing: usage and top-1 share are class-prior-weighted over the last 10 epochs, and NMI and assignment similarity use the last checkpoint. The lowest first-choice share is reported because usage alone does not reveal an expert selected mainly as the second choice. The last four rows measure expert function at the selected checkpoint; bold marks the strongest specialization.

TABLE S13
FUNCTIONAL MEASURES OF THE 12 MOE RUNS OF THE MAIN EXPERIMENTS, PART 1: ACCURACY-BASED MEASURES IN PP (VALIDATION SET, SELECTED CHECKPOINT; SEEDS 3 / 4 / 5).

Method	**Val. A_π (%)**	d_1	**Routing gain**	**Spec. gain**	**Top-1-only drop**
Fixed MoE	68.33 / 70.18 / 69.64	+0.30 / +0.23 / +0.74	+0.85 / +0.32 / +0.48	+1.3 / +0.7 / +2.1	+0.71 / +0.04 / +0.72
Schedule MoE	69.65 / 70.07 / 68.85	−0.09 / +0.59 / −0.37	−0.06 / +0.17 / +0.35	+1.1 / +0.6 / +2.2	+0.47 / −0.08 / +0.45
Aux-free MoE	69.89 / 70.25 / 68.52	+6.56 / +6.42 / +5.18	+3.83 / +5.27 / +4.31	+13.4 / +17.5 / +16.6	+1.85 / +0.77 / +1.57
FuzzyAdp-MoE	68.98 / 69.88 / 69.18	+1.08 / +0.20 / +0.48	+0.76 / +0.43 / +0.54	+1.2 / +0.8 / +1.3	+0.40 / +0.43 / +0.31

TABLE S14
FUNCTIONAL MEASURES OF THE 12 MOE RUNS OF THE MAIN EXPERIMENTS, PART 2: SIMILARITY AND ASSIGNMENT MEASURES (VALIDATION SET, SELECTED CHECKPOINT; SEEDS 3 / 4 / 5).

Method	CKA	Disagr.	Agreement	NMI	Any-of-5
Fixed MoE	0.52 / 0.50 / 0.47	0.20 / 0.18 / 0.20	0.839 / 0.856 / 0.822	0.229 / 0.235 / 0.234	0.738 / 0.745 / 0.749
Schedule MoE	0.68 / 0.70 / 0.72	0.18 / 0.14 / 0.17	0.871 / 0.891 / 0.866	0.275 / 0.229 / 0.293	0.752 / 0.732 / 0.732
Aux-free MoE	0.32 / 0.25 / 0.32	0.40 / 0.45 / 0.46	0.633 / 0.637 / 0.607	0.241 / 0.242 / 0.216	0.802 / 0.787 / 0.796
FuzzyAdp-MoE	0.51 / 0.59 / 0.55	0.18 / 0.16 / 0.17	0.845 / 0.862 / 0.856	0.253 / 0.260 / 0.236	0.742 / 0.739 / 0.735

σ values were trained on the same machine. τ is not a specialization knob (0.5–1.0 pp at $\tau = 0.25$ and 1.0, the same as the center), and λ is a weak one in the direction reported in the literature ($\lambda = 0.5$: 2.3 pp; $\lambda = 4.5$: 0.8 pp). Accuracy does not follow specialization: the three-seed means are 70.10% at $\sigma = 0$, 70.11% at 0.1, 70.46% at 0.4, and 70.34% at 0.8, all within X. The top-1-only drop grows with specialization (1.9–3.0 pp at $\sigma = 0$, 1.1–1.6 pp at 0.1, 0.3–0.4 pp at 0.4). NMI stays at 0.20–0.28 at every level of specialization.

Two controller runs show when the noise acts. Schedule MoE ends with $\sigma = 0.1$, yet its experts are as similar as those of Fixed MoE at $\sigma = 0.8$, not 0.1; FuzzyAdp-MoE holds $\sigma_0 = 0.4$ during its early mode and ends at the level of Fixed MoE at $\sigma = 0.4$. The class-to-expert assignment of Fixed MoE settles by epochs 21–26 while the learning rate is still high; the noise before that point sets the specialization, and lowering it afterwards does not change the experts.

D. Winner-Take-All Loss

The last rows of Tables S15 and S16 are a pilot of a different classification loss, reported in the future-work paragraph of the main paper. All main experiments compute the cross-entropy on the mixed prediction of the two selected experts, so the experts are trained to complement each other. The winner-take-all loss instead takes, for each image, the cross-entropy of the selected expert with the lower loss, and adds $-\log w$ of that expert so that the router learns to prefer it. Only the winning expert receives a gradient from the image. The runs use Fixed MoE with $(\lambda, \sigma, \tau) = (1.5, 0, 0.5)$, the three pilot seeds, and otherwise the settings of the main experiments.

The experts become mutually exclusive. The two selected experts predict the same class on 4.9–5.3% of the validation images, against 50–54% with the mixed loss at $\sigma = 0$ and 83–85% for Fixed MoE at $\sigma = 0.4$. Masking the top-1 expert costs 52.7–53.9 pp, and at least one of the five experts is correct on 86–87% of the images. Balance does not need a controller: the lowest expert usage over the last 10 epochs is 12.9–15.4%, and no epoch of any run falls below 5%. The mean validation A_π is 70.24%, which is -0.22 pp from Fixed MoE at $\sigma = 0.4$ (70.46%) and within X. Running only the top-1 expert costs 0.22, 0.27, and 0.28 pp on the three seeds. With the mixed loss at $\sigma = 0$ the same variant costs 1.9–3.0 pp, so specialization from low noise alone does not allow top-1 deployment, while this loss does. These runs are on the pilot seeds and the validation set only; they were not evaluated on the test split.

S5. PER-CLASS RESULTS

Table S17 gives the test results of Table II of the main paper with the range over the three seeds and the range of the seed-paired differences to Fixed MoE. At the last epoch instead of the checkpoint with the best validation A_π, the seed-paired gain of FuzzyAdp-MoE over Fixed MoE is $+0.77$ pp. The training subset is small, 41,130 images or 6.1% of the per-day training split, yet it keeps most of the accuracy of full-split training: the monolithic EfficientNet-B0 reaches 65.47% overall accuracy, 97.5% of the 67.17% that Gragnaniello et al. report for the same backbone trained on the full per-day training split (cited in the main paper), with about 16 times less training data.

Table S18 gives the per-class recall and F1 of the five models on the per-day test split (59,793 images), averaged over the three seeds. The class structure dominates the differences between models. Every model has F1 below 0.6 on 21 of the 27 classes. The four largest smooth or snow classes (dry asphalt smooth, dry concrete smooth, wet asphalt smooth, fresh snow) reach 81–92% recall for every model. The unevenness levels are the hard classes: the six *slight* and six *severe* classes reach 2–66% recall, with wet asphalt severe at 1.5–3.7% and water asphalt slight at 12–18%. FuzzyAdp-MoE has the highest mean recall on 11 of the 27 classes, Schedule MoE on 7, the monolithic network and Aux-free MoE on 4 each, and Fixed MoE on 1; the per-class differences between models are small against the differences between classes and were not tested for significance.

S6. EDGE MEASUREMENT DETAILS

A. Boards and Software

Table S19 lists the four boards. All measurements are CPU-only with TorchScript models at batch size 1, with the images loaded in the main process, one inter-op thread, and five warm-up forward passes on random inputs. The thread count equals the number of physical cores. The boards run the same software stack (Python 3.13.5, PyTorch 2.11.0+cpu, torchvision 0.26.0+cpu, timm 1.0.22, NumPy 2.2.6, Pillow 12.0.0), so that the only difference between boards is the hardware; PyTorch 2.10.0, the version of the training machine, was not usable on the Raspberry Pi 4B (its aarch64 wheel uses ARMv8.1 atomic instructions). The Jetson boards use their stock L4T kernels without `jetson_clocks`. The Xavier NX runs in `MODE_15W_6CORE`, because its factory default mode enables only four of its six cores. The frequency governor

TABLE S15
FUNCTIONAL MEASURES OF THE PILOT RUNS, PART 1: ACCURACY-BASED MEASURES IN PP (VALIDATION SET, SELECTED CHECKPOINT; SEEDS 0 / 1 / 2 UNLESS A SINGLE SEED IS GIVEN). FIXED MOE USES $(\lambda, \sigma, \tau) = (1.5, \sigma, 0.5)$ UNLESS STATED.

Run	**Val. A_π (%)**	d_1	**Routing gain**	**Spec. gain**	**Top-1-only drop**
Fixed, $\sigma = 0$	69.73 / 70.25 / 70.31	+12.71 / +13.31 / +15.23	+9.94 / +10.50 / +11.08	+20.4 / +24.1 / +27.4	+1.92 / +3.03 / +2.24
Fixed, $\sigma = 0.1$	69.51 / 70.79 / 70.04	+4.47 / +6.69 / +7.36	+4.26 / +6.00 / +5.45	+12.9 / +17.7 / +19.9	+1.56 / +1.10 / +1.27
Fixed, $\sigma = 0.2$, seed 0	69.64	+0.93	+1.49	+3.1	+0.86
Fixed, $\sigma = 0.2$, $\tau = 0.25$, seed 0	70.18	+0.51	+1.00	+5.5	+0.10
Fixed, $\sigma = 0.2$, $\tau = 1.0$, seed 0	69.00	+1.03	+0.85	+3.0	+1.00
Fixed, $\sigma = 0.2$, $\lambda = 0.5$, seed 0	70.25	+2.29	+1.83	+7.3	+0.73
Fixed, $\sigma = 0.2$, $\lambda = 4.5$, seed 0	69.70	+0.80	+0.67	+5.6	+0.55
Fixed, $\sigma = 0.4$	70.11 / 70.82 / 70.46	+0.42 / +0.54 / +1.05	+0.56 / +0.69 / +0.85	+1.8 / +1.7 / +2.3	+0.32 / +0.44 / +0.34
Fixed, $\sigma = 0.8$	70.43 / 70.26 / 70.34	+0.56 / +0.19 / +0.38	+0.75 / +0.61 / +0.51	+1.1 / +0.3 / +1.5	+0.16 / +0.61 / +0.40
Aux-free ($\sigma = 0$)	69.54 / 69.72 / 70.40	+5.64 / +4.11 / +3.81	+4.42 / +3.74 / +3.81	+13.2 / +14.4 / +12.6	+2.62 / +0.76 / +2.19
Schedule (σ 0.8 $\rightarrow$ 0.1), seed 0	70.23	+0.27	+0.30	+1.3	+0.19
FuzzyAdp (round 3)	69.53 / 71.38 / 69.74	+0.15 / +0.72 / +0.05	+0.61 / +0.49 / −0.24	+1.9 / +0.9 / +1.0	+0.55 / +0.41 / +0.66
Winner-take-all loss, $\sigma = 0$	69.54 / 71.27 / 69.91	+52.65 / +53.86 / +53.24	+32.59 / +33.35 / +32.48	+71.7 / +73.3 / +72.6	+0.22 / +0.27 / +0.28

TABLE S16
FUNCTIONAL MEASURES OF THE PILOT RUNS, PART 2: SIMILARITY AND ASSIGNMENT MEASURES (SAME RUNS AS TABLE S15).

Run	**CKA**	**Disagr.**	**Agreement**	**NMI**	**Any-of-5**
Fixed, $\sigma = 0$	0.34 / 0.29 / 0.28	0.52 / 0.56 / 0.57	0.537 / 0.496 / 0.504	0.216 / 0.200 / 0.225	0.807 / 0.801 / 0.804
Fixed, $\sigma = 0.1$	0.31 / 0.28 / 0.25	0.41 / 0.45 / 0.49	0.672 / 0.619 / 0.609	0.235 / 0.210 / 0.229	0.799 / 0.793 / 0.797
Fixed, $\sigma = 0.2$, seed 0	0.37	0.30	0.767	0.241	0.783
Fixed, $\sigma = 0.2$, $\tau = 0.25$, seed 0	0.49	0.28	0.790	0.234	0.785
Fixed, $\sigma = 0.2$, $\tau = 1.0$, seed 0	0.31	0.30	0.736	0.211	0.782
Fixed, $\sigma = 0.2$, $\lambda = 0.5$, seed 0	0.42	0.33	0.732	0.231	0.792
Fixed, $\sigma = 0.2$, $\lambda = 4.5$, seed 0	0.47	0.27	0.771	0.203	0.781
Fixed, $\sigma = 0.4$	0.56 / 0.59 / 0.47	0.18 / 0.19 / 0.19	0.854 / 0.833 / 0.845	0.267 / 0.219 / 0.261	0.744 / 0.749 / 0.751
Fixed, $\sigma = 0.8$	0.75 / 0.72 / 0.71	0.11 / 0.15 / 0.13	0.903 / 0.880 / 0.899	0.276 / 0.242 / 0.281	0.708 / 0.727 / 0.720
Aux-free ($\sigma = 0$)	0.24 / 0.27 / 0.28	0.44 / 0.41 / 0.40	0.613 / 0.646 / 0.674	0.242 / 0.195 / 0.230	0.804 / 0.787 / 0.781
Schedule (σ 0.8 $\rightarrow$ 0.1), seed 0	0.68	0.14	0.893	0.239	0.722
FuzzyAdp (round 3)	0.59 / 0.60 / 0.49	0.18 / 0.16 / 0.17	0.856 / 0.851 / 0.849	0.274 / 0.226 / 0.266	0.740 / 0.743 / 0.744
Winner-take-all loss, $\sigma = 0$	0.37 / 0.31 / 0.36	0.71 / 0.69 / 0.72	0.053 / 0.049 / 0.049	0.126 / 0.119 / 0.114	0.864 / 0.871 / 0.861

TABLE S17
TEST RESULTS OF THE FIVE MODELS, MEAN (MIN–MAX) OVER THREE SEEDS (PER-DAY TEST SPLIT).

Model	A_π **(%)**	**Acc. (%)**	**BA (%)**	**Macro-F1**	ΔA_π **vs. Fixed MoE (pp)**
Monolithic EfficientNet-B0	58.73 (57.71–59.46)	65.47	44.47	0.4140	+0.03 (−0.91–+0.51)
Fixed MoE	58.70 (57.20–60.36)	65.25	44.40	0.4114	–
Schedule MoE	58.52 (57.14–60.32)	64.67	44.68	0.4085	−0.18 (−0.41–−0.05)
Aux-free MoE	58.22 (57.48–59.52)	65.03	44.02	0.4071	−0.49 (−2.88–+0.98)
FuzzyAdp-MoE	**59.31** (57.87–60.22)	**65.80**	**45.24**	**0.4197**	**+0.61** (−0.15–+1.30)

Note: ΔA_π is the seed-paired difference to Fixed MoE, mean (min–max). Differences below $X = 0.50$ pp are treated as ties (Sec. IV-C.4 of the main paper). The best value in each column is in bold.

TABLE S18
PER-CLASS TEST RECALL (%) AND F1 OF THE FIVE MODELS (MEAN OVER THREE SEEDS; PER-DAY TEST SPLIT). M = MONOLITHIC, FX = FIXED MOE, SC = SCHEDULE MOE, AF = AUX-FREE MOE, FZ = FUZZYADP-MOE.

		Recall (%)					F1				
Class	**Images**	**M**	**Fx**	**Sc**	**Af**	**Fz**	**M**	**Fx**	**Sc**	**Af**	**Fz**
dry asphalt severe	332	36.2	39.5	35.8	42.1	35.5	0.267	0.285	0.281	0.280	0.285
dry asphalt slight	4404	34.8	34.5	35.5	35.2	32.7	0.461	0.458	0.465	0.461	0.444
dry asphalt smooth	10575	89.9	88.3	85.8	89.4	90.4	0.834	0.829	0.826	0.832	0.833
dry concrete severe	1129	56.5	54.2	52.0	50.9	54.8	0.474	0.438	0.427	0.427	0.452
dry concrete slight	4170	33.0	34.1	31.4	31.5	36.8	0.422	0.439	0.410	0.413	0.454
dry concrete smooth	11063	84.7	85.6	86.1	85.3	84.7	0.853	0.851	0.851	0.856	0.854
dry gravel	1657	53.8	54.0	53.1	51.0	51.1	0.367	0.382	0.363	0.361	0.374
dry mud	309	18.9	20.9	14.7	12.0	25.5	0.197	0.231	0.170	0.143	0.264
fresh snow	3731	91.7	91.3	90.3	90.5	91.2	0.949	0.947	0.940	0.943	0.946
ice	363	62.2	61.5	63.5	60.8	68.6	0.386	0.396	0.376	0.389	0.413
melted snow	413	19.8	18.9	21.1	22.4	16.1	0.250	0.245	0.257	0.269	0.228
water asphalt severe	273	27.4	23.6	26.5	28.4	29.1	0.316	0.300	0.324	0.340	0.347
water asphalt slight	115	12.2	14.5	14.8	16.8	17.7	0.061	0.072	0.075	0.076	0.076
water asphalt smooth	793	56.2	56.8	58.8	59.9	55.7	0.271	0.279	0.280	0.277	0.290
water concrete severe	416	20.1	24.8	27.6	16.2	22.9	0.196	0.201	0.220	0.168	0.197
water concrete slight	761	24.2	27.6	25.8	25.2	29.0	0.275	0.276	0.267	0.267	0.277
water concrete smooth	3293	40.9	40.0	41.3	37.8	41.3	0.406	0.411	0.413	0.399	0.421
water gravel	1646	32.5	29.1	27.5	32.5	30.7	0.454	0.415	0.390	0.442	0.430
water mud	582	34.9	34.7	37.7	35.0	32.2	0.276	0.266	0.291	0.257	0.275
wet asphalt severe	286	2.4	2.6	1.5	2.8	3.7	0.041	0.042	0.026	0.046	0.060
wet asphalt slight	737	68.0	67.0	67.7	64.4	64.5	0.514	0.514	0.496	0.506	0.505
wet asphalt smooth	6315	81.6	81.2	79.1	81.8	81.8	0.854	0.840	0.837	0.846	0.849
wet concrete severe	77	8.2	7.8	14.3	8.2	13.0	0.084	0.068	0.106	0.077	0.116
wet concrete slight	102	28.8	21.2	25.8	22.9	22.2	0.171	0.109	0.136	0.139	0.112
wet concrete smooth	3585	42.6	41.8	43.0	40.9	43.5	0.478	0.471	0.487	0.467	0.491
wet gravel	1589	63.2	61.8	65.5	63.3	63.1	0.659	0.666	0.661	0.656	0.667
wet mud	1077	76.2	81.6	80.1	81.3	83.5	0.661	0.676	0.656	0.656	0.672

is the system default on every board (frequencies are not pinned). Every result file records the governor, frequencies, temperatures, power mode, and package versions at the time of measurement.

B. Models and Consistency Check

Six models were exported from the seed-3 checkpoints of the main experiments on each board with `torch.jit.script`: the monolithic network, the four MoE variants with top-2 routing, and Fixed MoE with top-1 routing (only the first-choice expert is run, with weight 1). At export, the scripted model was checked against the eager model on 8 random inputs and 32 real images (identical argmax, maximum logit difference below 10^{-3}). Reference predictions for the 5,810 validation images were produced on a laptop CPU in fp32 with the frozen evaluation code. On every board, the five top-2 models agree with the reference on all 5,810 images (agreement 1.0000) and select the same top-2 expert set for every image; the acceptance threshold was 0.999. The top-1 variant agrees with the top-2 reference on 96.28% of the images on every board, and its first-choice expert equals the reference's first choice on every image, so the routing is unchanged and the difference comes only from dropping the second expert. The validation A_π measured on the boards (monolithic 64.39%, Fixed MoE top-2 64.04%, top-1 63.37%) is the same on all four boards.

C. Latency

Latency was measured over the 5,810 validation images, one image at a time, recording the model time (the forward pass only) and the end-to-end time (including image loading and preprocessing) for every image. Table S20 reports the mean, median, and 90th percentile of the model time, the end-to-end time per image defined as in the main paper's latency table (loop time divided by the number of images), and the throughput.

Checks. On the Raspberry Pi boards, the under-voltage and throttling flag (`vcgencmd get_throttled`) was `0x0` before and after every measurement, and the final temperatures were 54–55 °C (4B) and 58–60 °C (5), below the 80 °C throttling point. The Raspberry Pi 5 was powered from the official 5.1 V / 3 A supply of the 4B after its original supply showed under-voltage events under load; with the 3 A supply, a 10-minute full load produced no under-voltage event. The Jetson boards do not expose such a flag. The four top-2 variants differ by less than 1% on the Raspberry Pi boards and by 1.3–1.5% on the Jetson boards; the controller affects training only.

TABLE S19
BOARDS USED FOR THE EDGE MEASUREMENTS.

Board	CPU	Threads	OS	Governor	Power mode
Raspberry Pi 4B	4× Cortex-A72, 1.8 GHz	4	Debian 13	ondemand	default
Raspberry Pi 5	4× Cortex-A76, 2.4 GHz	4	Debian 13	ondemand	default
Jetson Xavier NX	6× Carmel, 1.42 GHz	6	Ubuntu 20.04 (L4T R35.6)	schedutil	`MODE_15W_6CORE`
Jetson Orin Nano	6× Cortex-A78AE, 1.34 GHz	6	Ubuntu 22.04 (L4T R36.4)	schedutil	25 W (default)

TABLE S20
LATENCY PER IMAGE (MS) AND THROUGHPUT ON THE FOUR BOARDS (5,810 VALIDATION IMAGES, BATCH 1, CPU-ONLY TORCHSCRIPT). RATIO = MEAN MODEL TIME RELATIVE TO THE MONOLITHIC NETWORK ON THE SAME BOARD.

Board	Model	Model mean	Median	p90	Std.	End-to-end	img/s	Ratio
RPi 4B	Monolithic	227.45	226.84	231.09	2.32	236.67	4.23	1.000
	Fixed MoE, top-2	241.31	241.04	243.11	1.34	248.57	4.02	1.061
	Schedule MoE, top-2	241.80	241.57	243.79	1.47	248.98	4.02	1.063
	Aux-free MoE, top-2	242.07	241.79	243.93	1.34	249.30	4.01	1.064
	FuzzyAdp-MoE, top-2	241.31	240.86	242.95	6.78	248.50	4.02	1.061
	Fixed MoE, top-1	229.54	229.23	231.35	1.33	236.77	4.22	1.009
RPi 5	Monolithic	76.33	76.41	77.07	0.61	80.84	12.37	1.000
	Fixed MoE, top-2	82.04	82.03	82.65	0.85	85.46	11.70	1.075
	Schedule MoE, top-2	82.37	82.45	83.01	0.66	85.80	11.66	1.079
	Aux-free MoE, top-2	81.94	82.00	82.60	0.72	85.36	11.71	1.073
	FuzzyAdp-MoE, top-2	81.84	81.90	82.47	0.53	85.25	11.73	1.072
	Fixed MoE, top-1	77.21	77.22	77.94	0.68	80.64	12.40	1.012
Xavier NX	Monolithic	93.47	90.85	99.94	11.41	100.90	9.91	1.000
	Fixed MoE, top-2	98.53	97.42	100.87	8.27	106.04	9.43	1.054
	Schedule MoE, top-2	99.00	97.96	101.34	8.35	106.43	9.40	1.059
	Aux-free MoE, top-2	98.23	97.36	100.08	8.60	105.67	9.46	1.051
	FuzzyAdp-MoE, top-2	99.53	98.50	101.07	7.36	106.96	9.35	1.065
	Fixed MoE, top-1	92.32	90.96	95.49	8.09	99.89	10.01	0.988
Orin Nano	Monolithic	80.57	75.12	83.07	20.11	86.63	11.54	1.000
	Fixed MoE, top-2	87.78	81.18	93.45	22.65	93.79	10.66	1.089
	Schedule MoE, top-2	87.49	81.15	91.15	22.45	93.54	10.69	1.086
	Aux-free MoE, top-2	87.49	81.27	90.90	21.90	93.58	10.69	1.086
	FuzzyAdp-MoE, top-2	88.77	82.98	90.62	21.87	94.95	10.53	1.102
	Fixed MoE, top-1	82.32	76.23	86.08	21.42	88.39	11.31	1.022

Two anomalies, kept as measured. The first monolithic run on the Xavier NX was the first measurement after a reboot into the new power mode: its model time fell from 111 ms to about 95 ms over the first 2,000 images (standard deviation 21 ms), which would have made the top-1 variant 5.7% faster than the monolithic network. The monolithic model was re-measured after the other five models; the table reports the re-measurement. A re-measurement of the top-1 variant immediately afterwards reproduced the first one within 0.6%. On the Orin Nano, the model-time standard deviation is 20–23 ms and the mean exceeds the median by about 5.5 ms for every model: a fraction of the images is markedly slower, while the running mean over blocks of 500 images is flat (79–82 ms for the monolithic network). The cause was not identified; the means include these images.

D. Power and Energy

Raspberry Pi (discharge method). Each board is powered only by a power bank charged to full, runs inference continuously until the bank shuts off, and writes a heartbeat every 10 s with the image count and the throttling flag. Energy per image is $E = E_{\text{out}}/N$ and mean power is $P = E_{\text{out}}/T$, where N is the number of images processed and T the run time. The manufacturers do not state the 5 V output capacity, so E_{out} is reported as $37.0\ \text{Wh} \times [0.70, 0.85]$ for both banks (nominal cell energy 3.7 V × 10,000 mAh). The lower bound follows the Intenso manual, which states that about 30% of the capacity is lost in conversion; the upper bound is a typical measured efficiency of 5 V boost converters and not a manufacturer value. The energy ratio of two models on the same board and bank, $E_{\text{MoE}}/E_{\text{single}} = N_{\text{single}}/N_{\text{MoE}}$, does not depend on E_{out}. The RPi 4B uses an Intenso F10000 and the RPi 5 a Goji G6P10BK25; each board keeps its own bank, and absolute

values are not compared across boards. A run is valid if the throttling flag first becomes non-zero, if at all, within the last 5% of the run time, which is the normal voltage drop of an emptying bank. Each board measures the monolithic network and Fixed MoE top-2 twice, interleaved. Over the two pairs, the energy ratio of Fixed MoE top-2 to the monolithic network is 1.06 on the RPi 4B (1.05 and 1.07) and 1.04 on the RPi 5 (1.01 and 1.08).

In the first RPi 4B pair (Table S21), both runs were valid: the flag stayed `0x0` to the end and the bank shut off without an earlier under-voltage event. The energy ratio is 1.071, which decomposes into the throughput ratio 1.055 and the mean-power ratio 1.015 (the MoE run was 1.5% shorter on the same energy). The throughput during discharge is within 1% of the throughput of the latency measurement. The second monolithic run on the RPi 4B reproduced the first within 1.1% in N and 0.8% in T. In the first RPi 5 pair, both runs were valid: the flag stayed `0x0` and the input voltage stayed at 4.95–5.02 V to the last heartbeat. Its energy ratio is 1.010. The monolithic run processed 8.3% more images per second, but the MoE run lasted 7.2% longer on the same bank energy, so its mean power was 6.7% lower. The CPU temperature does not explain this difference (58.5 °C for the MoE run against about 57 °C for most of the monolithic run), and it did not repeat in the second pair, where the mean power of the two models differs by 0.7%. One further RPi 4B run was started on the bank assigned to the RPi 5 and was discarded before analysis. The second pairs give energy ratios of 1.051 on the RPi 4B and 1.078 on the RPi 5. The main paper reports the ratio of the two-run means, 1.06 on the RPi 4B and 1.04 on the RPi 5.

Jetson (on-board power monitor). The input power `VDD_IN` of the on-board INA3221 monitor is logged by `tegrastats` at 1 s intervals in the background, first for 2 min at idle and then for 20 min of continuous inference; the inference loop never reads the monitor. Only the board input rail is used, because the other rails are its components. The logged value is the instantaneous reading, not the moving average that `tegrastats` also prints. On the Xavier NX, the rail names are readable only by root, so its power runs are executed as root. The energy per image is P_{load} times the end-to-end time per image of the same run; $(P_{\text{load}} - P_{\text{idle}})$ times that time is reported as the incremental energy. The acceptance criterion is an end-to-end time within 2% of the latency measurement of the same model, which bounds the overhead of the logger. Each board measures the monolithic network, Fixed MoE top-2, and Fixed MoE top-1 once; the standard deviation over the 20 min gives the fluctuation.

Orin Nano. All three runs meet the 2% criterion (Table S22). The load power of the three models differs by less than 0.5%, so the energy differences come from the time per image. The energy ratio to the monolithic network is 1.071 for top-2 routing (1.067 on the incremental energy) and 1.010 for the top-1 variant (1.004).

Xavier NX. In a first round, the three power runs were 2.9%, 6.4%, and 3.3% slower than the latency measurements of the same models and failed the criterion; that round is not used. A diagnosis on the monolithic model showed that the background logger slows inference by about 1% and that running as root has no effect, and the cores stayed at 1.40–1.42 GHz with a maximum temperature of 35 °C, so neither the logger nor throttling explains the first round. The cause was not identified. The three runs were repeated; the table reports the repetition. Its top-2 run exceeds the criterion by 0.24 percentage points, and the other two meet it. The three repeated runs are used together, because they were measured consecutively under the same conditions and the model time of the monolithic network on this board varied by about 8% across six measurements. The energy ratio to the monolithic network is 1.063 for top-2 routing (1.049 on the incremental energy) and 0.986 for the top-1 variant; the first round gave 1.053 and 0.991. The absolute energies of the two rounds differ by 2–4%.

E. Scope of the Measurements

The discharge method gives reliable ratios; its absolute values carry the capacity uncertainty of the banks (about $\pm 10\%$) and are not compared across boards or with the Jetson values. Latency was measured on the validation images with the seed-3 checkpoints; neither the images nor the weights affect the timing. Thermal behavior was recorded, not controlled, and the boards run their default governors, so the numbers describe the default configuration of each board rather than its best configuration. The Raspberry Pi 5 was measured with a 3 A supply instead of its 5 A supply.

S7. Reproducibility

Environment. All training and evaluation ran on one machine with an NVIDIA GeForce RTX 4070 Laptop GPU (driver 595.91.07), Python 3.12.12, PyTorch 2.10.0+cu128, torchvision 0.25.0, timm 1.0.22, NumPy 2.2.6, Pillow 12.0.0, scikit-fuzzy 0.5.0, CUDA 12.8, and cuDNN 91002. These versions are stored in `env_lock.json`; the training and evaluation scripts compare the running environment against it and refuse to start on a mismatch. Every run writes its environment and the git commit of the code into its `config.json`.

Determinism. Training uses deterministic kernels and fixed seeds. A repeated run of the same configuration reproduces the original bit for bit: all per-epoch values of the history, the evaluation metrics, and every per-image prediction. This was verified on three seeds under an earlier version of the code; the later code changes were checked in short runs to leave the training arithmetic of Fixed MoE unchanged. The training diagnostics that produce the routing statistics do not change the training arithmetic (verified bit for bit with the diagnostics on and off).

Training settings. All models are trained with AdamW for a fixed budget of 50 epochs without early stopping, so every variant receives the same number of optimization steps. The learning rate peaks at 10^{-3} after 3 warm-up epochs and decays to 10^{-6} with a cosine schedule. The batch size is 64, training uses mixed precision, and the inputs are 224×224. Training augmentation consists of horizontal flips, scaling by 0.9–1.1, and mild color jitter. These settings are identical across all

TABLE S21
RASPBERRY PI DISCHARGE RUNS (4 THREADS). ENERGY AND POWER ARE GIVEN AT THE TWO BOUNDS OF E_{OUT}.

Board	Model	Run	T (h)	N	img/s	E (J/image)	P (W)
RPi 4B	Monolithic	1	7.27	111,431	4.26	0.837–1.016	3.56–4.33
RPi 4B	Fixed MoE, top-2	1	7.16	104,089	4.04	0.896–1.088	3.61–4.39
RPi 4B	Monolithic	2	7.21	110,167	4.25	0.846–1.028	3.59–4.36
RPi 4B	Fixed MoE, top-2	2	7.27	104,787	4.00	0.890–1.080	3.56–4.32
RPi 5	Monolithic	1	4.72	211,842	12.47	0.440–0.534	5.49–6.67
RPi 5	Fixed MoE, top-2	1	5.06	209,711	11.52	0.445–0.540	5.12–6.22
RPi 5	Monolithic	2	5.08	227,146	12.43	0.410–0.498	5.10–6.19
RPi 5	Fixed MoE, top-2	2	5.04	210,806	11.61	0.442–0.537	5.14–6.24

TABLE S22
JETSON POWER RUNS (6 THREADS; `VDD_IN` FROM THE INA3221 MONITOR, 1 S SAMPLING, 2 MIN IDLE AND 20 MIN LOAD).

Board	Model	P_{idle} (W)	P_{load} (W)	End-to-end vs. latency run	E (J/image)	ΔE (J/image)
Xavier NX	Monolithic	1.71	5.05	$+1.27\%$	0.516	0.341
Xavier NX	Fixed MoE, top-2	1.75	5.06	$+2.24\%$	0.548	0.358
Xavier NX	Fixed MoE, top-1	1.74	5.20	-1.95%	0.509	0.339
Orin Nano	Monolithic	2.92	5.30	$+0.6\%$	0.461	0.207
Orin Nano	Fixed MoE, top-2	2.92	5.28	-0.2%	0.494	0.221
Orin Nano	Fixed MoE, top-1	2.92	5.27	$+0.0\%$	0.466	0.208

methods except the weight decay, which is 10^{-4} for the MoE variants and 0.01 for the monolithic model, each selected in the pilot study (Sec. S2).

Timing runs. Table S23 gives the training time of each model. The training times of the main paper come from 15 separate runs (five models, three seeds, one batch per seed) with the diagnostics off, no other task on the machine, and the display on. The accuracy columns of every timing run are bit-identical to those of the corresponding main run. The first batch of seed 3 ran with the display off: its MoE training was 4.1–6.0% slower per epoch than in the other two batches and validation was slower for all five models, while the monolithic training time was unchanged. Before repeating it, we wrote down the criterion that a display-on repetition must return to the level of the other batches (about 2.9 s validation for the monolithic model and 76–77 s per training epoch for the MoE models). The repetition gave 2.88 s and 76.5–76.9 s, and it replaces the first batch; the display-off batch is kept in the release and is not part of the reported times. With the display on, the total training time of a model varies by at most 0.4 min across the three seeds. The mechanism behind the slowdown was not identified. The steady-state time per epoch (mean from epoch 2, including validation) is 69.5 s for the monolithic model and 79.8–80.2 s for the four MoE variants, with a range of at most 0.5 s over the seeds.

Code version. The 33 training runs of the main experiments, the 66 test evaluations, and the 15 timing runs were produced from commit `411e887` of our development repository with a clean working tree. The pilot reruns of Sec. S4-D and the $\sigma = 0$ runs of seeds 1 and 2 were produced from commit `96521d2`, whose training code is unchanged for the mixed loss. The settings were frozen at commit `624f1b0` before the test split was used; the two commits differ only in a documentation file. The released code was exported from commit `7ffd0c4` of the same repository; from `96521d2` to `7ffd0c4` the code changed only in comments, docstrings, and the report text of three summary scripts (see the README of the release).

TABLE S23
TRAINING TIME (MIN; 50 EPOCHS, MEAN OF THREE SEEDS).

Model	Monolithic	Fixed	Schedule	Aux-free	FuzzyAdp
Time	58.3	67.7	67.7	68.0	67.5

Data. The RSCD per-day split (https://github.com/MiviaLab/Road-Surface-Dataset-per-day-split) provides the three manifests: train (678,281 images), validation pool (290,227), and test (59,793). The training subsets draw 4% of the full-dataset count of each class without replacement (41,130 images per subset, six subsets for seeds 0–5), and the validation subset draws $\max(0.5\%, 100)$ images per class from the validation pool (5,810 images, shared by all runs). Sampling sorts the paths within each class and uses NumPy's `default_rng` seeded by (seed, stream, class index), so the subsets are reproducible from the manifests alone. The class prior π is the class distribution of the training subsets. The SHA-256 checksums of the three source manifests and of every generated subset list are recorded in `manifest.json`. Seeds 0–2 were used for the pilot study only and seeds 3–5 for the main experiments; a seed selects both the training subset and the initialization.

Release. The code release contains the training, evaluation, dense-analysis, and summary scripts; the controller configuration files of the three design rounds; the subset manifests; the environment lock file; the per-run histories, configurations, and evaluation metrics of the main experiments; the device measurement scripts, raw result files, and board environment records; and this document.